

SwiftRoute Logistics — Business Questions Catalog

Every business question answered across the project, start to finish — dashboard and extended analysis combined

This is the master index of the SwiftRoute Logistics analytics project: every business question the project answers, in one place, organized by operational category rather than by tool. For each question it names where the answer lives (Power BI dashboard, Excel QA workbook, SQL notebook, or Python notebook) and summarizes the finding. Full workings, charts, and code sit in the source files — this document is the map.

A. Orders & Delivery Performance

#	Business Question	Answered In	Finding / Answer
1	What is total order volume for the selected period, and how does it compare month-over-month?	Power BI — Overview	1,127 orders in Sep 2024 (+1.8% MoM). Order volume overall is flat year-round (~1,150/month, no growth trend — confirmed in Python notebook Q9, $R^2 = 0.008$).
2	What % of orders are delivered on time, and how does it trend?	Power BI — Overview	79.3% on-time in Sep 2024 (−0.6% MoM). Baseline delay rate across the full dataset is ~20.4%.
3	What is overall Customer Satisfaction Score (CSAT), and how does it trend?	Power BI — Overview	84.1% in Sep 2024 (−0.4% MoM).
4	What is the average delivery time, and how does it trend?	Power BI — Overview	35.70 hours in Sep 2024 (−0.5% MoM).
5	What are the top delay reasons, and does the leading cause differ by hub?	Excel + SQL	All 10 delay reasons occur at a near-uniform ~9–11% frequency; the leading cause differs hub to hub with no consistent pattern — no single company-wide fix stands out.
6	What is the cancellation rate, and does it vary by hub, vehicle type, or weekday?	Excel + SQL	0.9% overall (252 of 27,979 orders); narrow range (0.6%–1.1%) across hubs and weekdays — not currently a material problem area.
7	Is the hub-to-hub on-time rate difference statistically real?	Python (chi-square)	$p = 0.174$ — not statistically significant. The dashboard ranking is real but the spread is within normal random variation.
8	Does vehicle type significantly affect delivery time?	Python (ANOVA)	$p = 0.58$ — not significant. Vehicle-type assignment doesn't need optimizing for speed.
9	Does a hub's processing time relate to that order's total delivery time?	Python (correlation)	$r \approx 0.00$, $p = 0.68$ — no relationship. Hub processing speed and end-to-end delivery time are functionally independent.
10	How much does a delay cost in CSAT points, and is it significant?	Python (t-test)	−1.56 points (2.94 vs 4.50), $p < 0.0001$ — the single strongest lever on customer satisfaction in the dataset.
11	What combination of factors best explains CSAT overall?	Python (multiple regression)	$R^2 = 0.54$; the delay flag dominates every other variable by more than an order of magnitude.
12	Can individual orders be flagged in advance as delay-risk?	Python (Random Forest)	AUC = 0.50 — no predictive signal in the fields available. A genuine negative finding: recommend capturing live traffic/weather/queue data before investing further.
13	Does the dashboard's MoM order growth figure reproduce independently outside Power BI?	SQL	Yes — SQL-native recomputation matches the DAX measure, confirming the dashboard figure.

B. Hub Operations

#	Business Question	Answered In	Finding / Answer
14	How many operational hubs does SwiftRoute have?	Power BI — Overview / Hubs	6 hubs.
15	How do orders processed compare to each hub's rated capacity?	Power BI — Hubs	Dallas Main Hub and Houston Hub carry the largest absolute volumes; Dallas Main Hub visibly exceeds its capacity line on the chart.
16	How do hubs rank on performance (on-time %)?	Power BI — Hubs	Ranked 83.3% (Houston) down to 76.7% (Fort Worth) — see Q7 above for why this spread isn't statistically significant.
17	How many hours does each hub take to process orders, by day of week?	Power BI — Hubs	Matrix view by hub × weekday; processing time is fairly stable across days for most hubs.
18	Which hubs run over their rated capacity, and is it a one-off spike or a consistent pattern?	Excel + SQL	Dallas Main Hub runs at ~121% of rated monthly capacity on average and is over 100% in nearly every month across both years — a structural mismatch, not a spike. Every other hub sits at 75–76%.
19	Is the capacity problem a growth problem, or a fixed structural one?	Python (trend analysis)	Structural — 2-year order volume is flat (no growth trend), so this is a one-time capacity/rebalancing fix, not an ongoing forecasting challenge.

C. Drivers & Workforce

#	Business Question	Answered In	Finding / Answer
20	How many active drivers does SwiftRoute have?	Power BI — Overview / Drivers	55 drivers.
21	How does driver experience relate to performance rating?	Power BI — Drivers (chart) + Python (statistical test)	Moderate, statistically significant positive correlation ($r = 0.45$, $p < 0.001$) — experience genuinely predicts a higher rating.
22	Which drivers have the most delays?	Power BI — Drivers	Ranked bar chart of delay share by driver, topped by John Moore (42.9%) in the sample period shown.
23	What does an individual driver's profile look like (hire date, experience, rating, monthly deliveries)?	Power BI — Drivers	Interactive driver-level card with a 12-month delivery trend line, selectable by name.
24	Does the 'Driver Name' recorded on each order actually match the correct driver?	Excel (QA)	No — order-level driver name text does not reliably match the driver master table for the same Driver ID. Every driver-related join in this project uses Driver ID as a result of this check.
25	How evenly is delivery workload spread across the 55 drivers?	Excel + SQL	Highly uneven: busiest driver (Christopher Martin) handled 1,281 deliveries over 2 years vs. 345 for the least-used (mean ≈ 504 , $\sim 3.7\times$ spread) — a dispatch/assignment fairness and efficiency finding.
26	Does employment type (full-time/part-time/contract) relate to on-time rate or CSAT?	SQL	No meaningful difference (79–82% on-time, CSAT within 0.02 points across types) — staffing-mix decisions can be made on cost/flexibility grounds alone.
27	Does driver tenure relate to delay rate?	SQL	Flat — 20.7% (0–2 yrs) to 21.5% (4+ yrs). Tenure alone isn't a lever on delay rate (even though it is on rating, per Q21).

D. Vehicles & Fleet

#	Business Question	Answered In	Finding / Answer
28	How large is the vehicle fleet?	Power BI — Overview / Vehicles	45 vehicles.
29	How many vehicles are currently active vs. in maintenance?	Power BI — Vehicles	33 active (73.3%), 12 in maintenance (26.7%).
30	How do order volumes compare across vehicle models?	Power BI — Vehicles	Freightliner M2 and Mercedes Sprinter handle the highest order counts among the models shown.
31	Does vehicle age relate to breakdown frequency?	Power BI (chart) + Python (statistical test)	Strong, significant correlation ($r = 0.65$, $p < 0.001$); age alone explains ~42% of breakdown variance — the best single predictor of breakdown risk available.
32	Which specific vehicles and models have the most breakdowns?	Power BI — Vehicles	Ranked bar charts by Vehicle Code and by Vehicle Model.
33	How are orders distributed across vehicle types?	Power BI — Vehicles	Van carries the majority share (62.7%), followed by Truck (23.4%).
34	Which vehicles are idle or under-utilized vs. overworked?	Excel + SQL	4 of 45 vehicles show zero linked deliveries over 2 years, while the busiest handled 1,263 — confirmed as a genuine utilization finding, not a data-linkage issue (100% of vehicle codes match the order log).
35	Does the current maintenance-alert field add predictive power beyond vehicle age alone?	Python (regression)	Barely — R^2 moves from 0.42 (age only) to 0.43 (age + alerts). Age is doing almost all the predictive work.
36	Can the fleet be segmented into an actionable maintenance risk tier?	Python (quantile segmentation)	Yes — oldest third of the fleet by age carries a disproportionate share of total breakdowns, giving fleet management a ready-made inspection priority list.

Project Coverage Summary

36 business questions in total: 17 answered directly on the Power BI dashboard for day-to-day stakeholder use, 6 answered in the Excel QA workbook as first-pass validated analysis, 8 answered in the SQL notebook using joins/window functions/trend analysis, and 10 answered in the Python notebook using statistical hypothesis testing, regression, classification, and segmentation (several questions are cross-referenced across more than one tool, e.g. a dashboard chart validated by a formal significance test). Roughly a third of the hypotheses tested — vehicle type effects, hub on-time differences, tenure-vs-delay, order-level delay prediction — turned out not to hold up statistically. Reporting those findings honestly alongside the ones that did is itself a deliverable: it tells SwiftRoute where not to spend its next improvement budget, not just where to.